SARS-CoV-2 Immuno-Vaccinology

Leonard Mada

Systematic Review: CoV-2

Contents:

- Overview CoV-2
- Literature: **in-silico** 2020 – 2025
- Immuno-Informatics:
  - Epitopes, Vaccines, Immuno-Simulations
- Pharmacology
  - Docking: Drugs, Phytocompounds

Overview: Human CoVs

Upper Respiratory Tract (URT):

- **Alpha-CoVs:** 229E, NL63;
- **Beta-CoVs:** OC43, KHU1;
- 15% of **common colds**: autumn/winter season;
- Usually **NO** pneumonia;

Lower Respiratory Tract (LRT):

- **Beta-CoVs:** SARS-CoV, SARS-CoV-2, MERS;

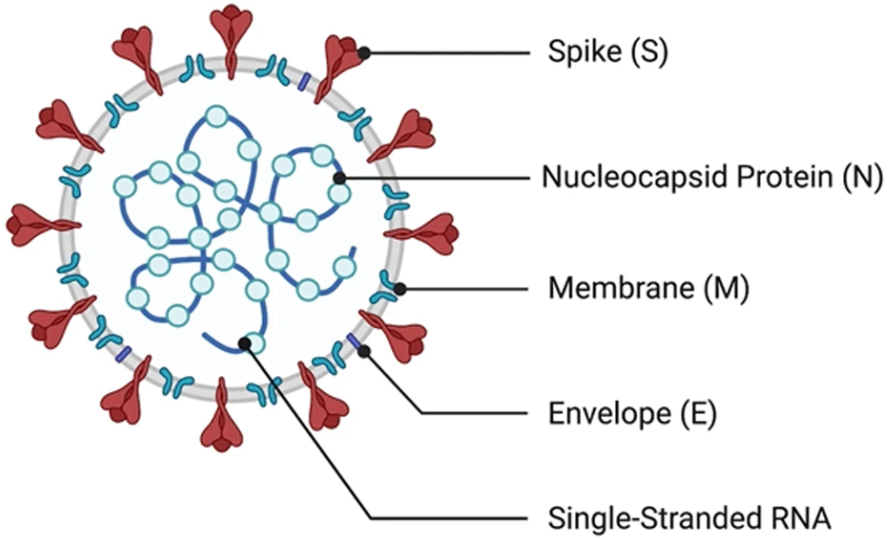
Overview: Human CoVs

Structure:

- EM: crown-like shape (corona);
- Enveloped;

Surface proteins:

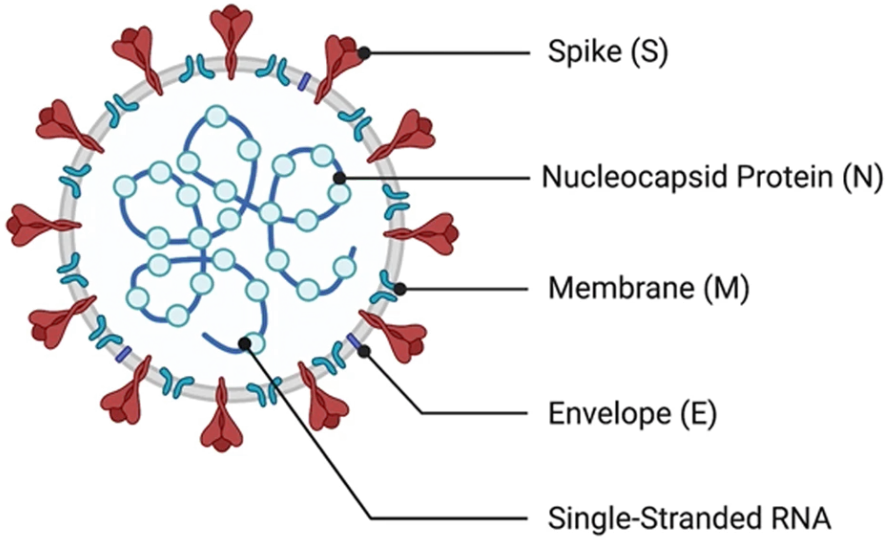
- **Spike: > 1000 AA;**
- **Membrane protein** (M);
- **Envelope** (E): < 100 AA;



Overview: Human CoVs

Host Receptor:

Strain	Receptor
229E	APN
NL63	ACE2
SARS CoV-1, 2	ACE2
MERS	DPP4



- **Spike: > 1000 AA;**

APN = Aminopeptidase N; DPP4 = Dipeptidyl peptidase 4;

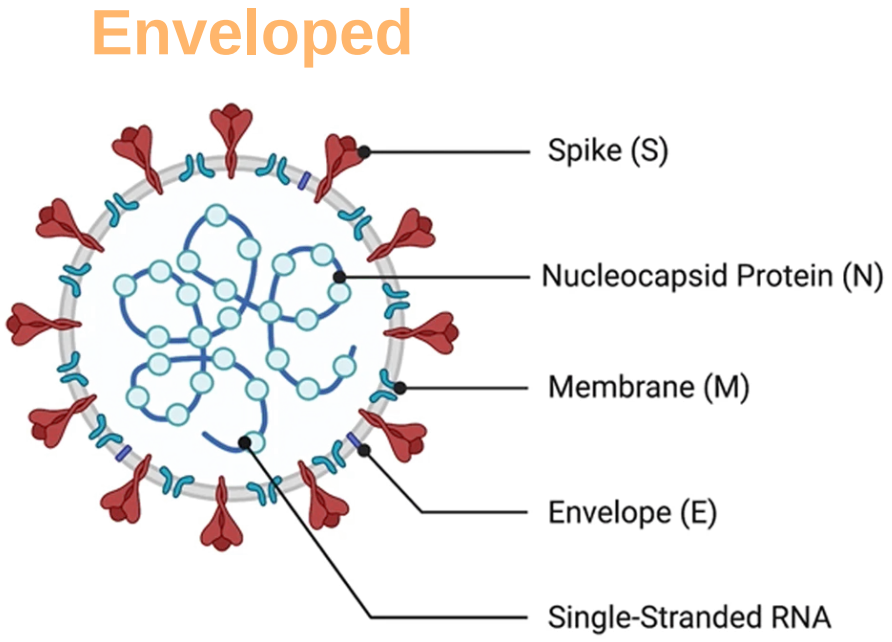
Overview: Human CoVs

Neutralising Antibodies:

- Only against: **S**, **M**, **E**; (*)
- **Cannot bind:** N, NSPs;

Surface proteins:

- **Membrane protein** (M);
- **Spike**: > 1000 AA;
- **Envelope** (E): < 100 AA;



* E may be too small (< 100 AA); also part of it is trans-membrane;

Overview: Human CoVs

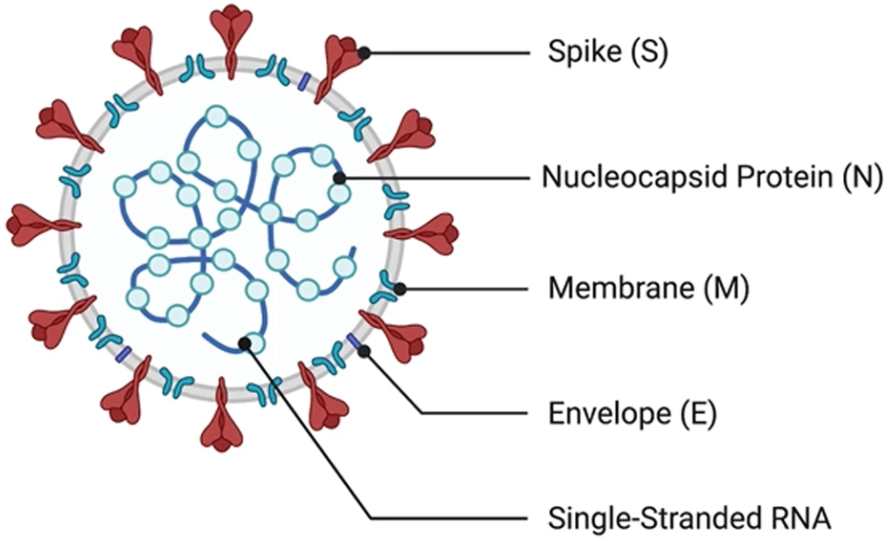
Neutralising Antibodies:

- Only against: **S**, **M**, **E**;
- **Cannot bind:** N, NSPs;

Anti-N Abs:

- Cell lysis => free N protein =>
  => Ig against **linear** N epitopes;
- **NO** neutralisation!
- May still be useful: Diagnostic assays; (#)

Enveloped



But NOT as vaccine that induces antibodies;

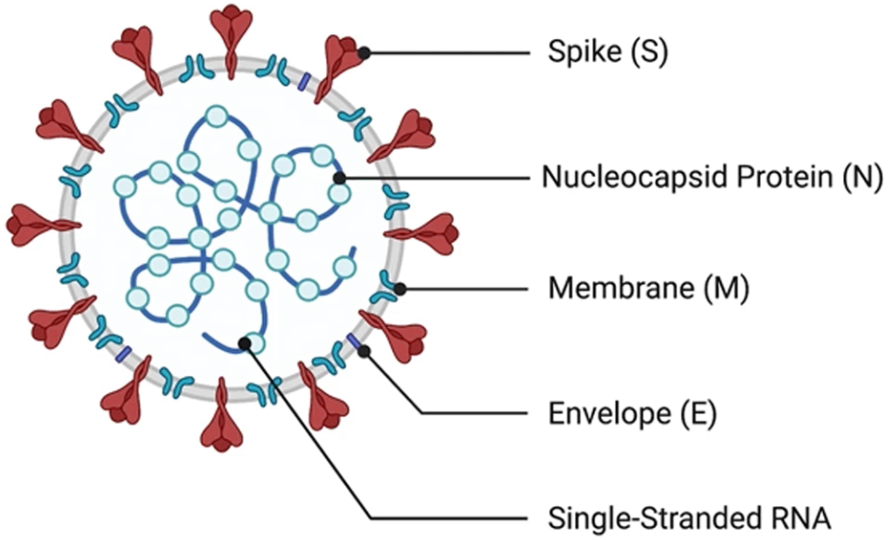
Overview: Human CoVs

Genome:

- Single stranded **RNA**;
- Plus-sense **RNA**;
- 30 kb (single molecule);

Genes:

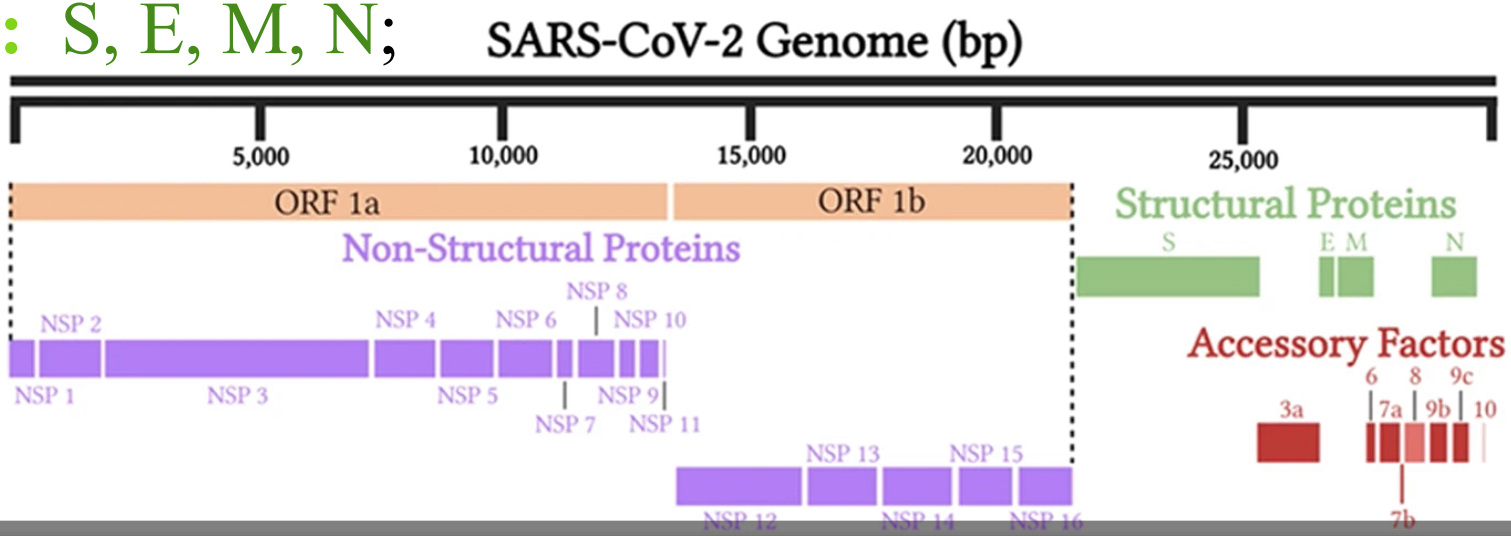
- NSPs & Structural proteins;



Overview: SARS CoV-2

Genome:

- **NSP:** ORF1a/b **Polyproteins**, others (Accessory);
- **SP:** S, E, M, N;

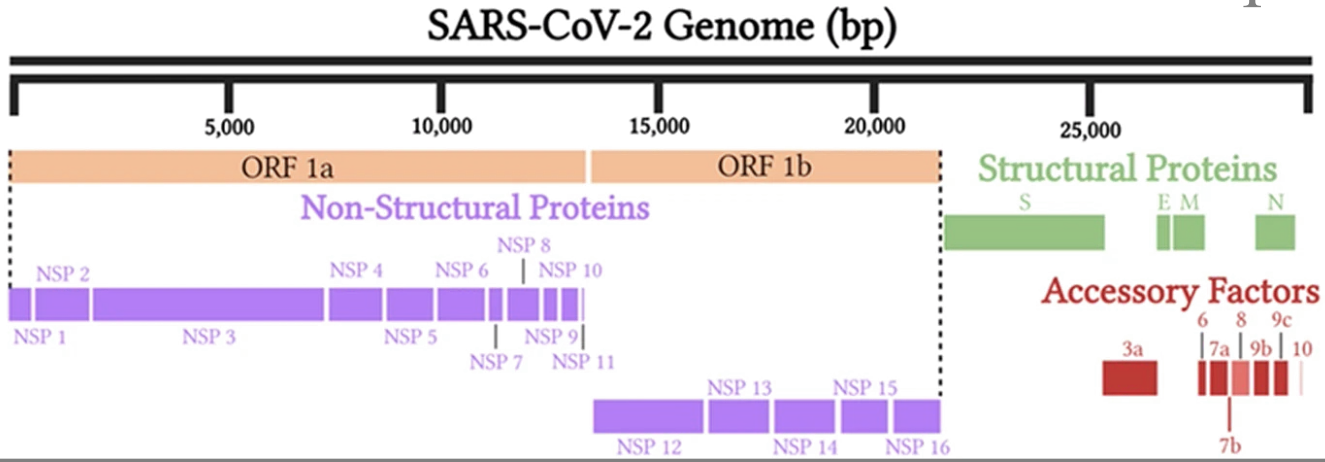


SP = Structural Proteins;

Overview: SARS CoV-2

Polyproteins: ORF1a/b

- Cleaved by **proteases**: into 10 & 5 proteins;



Proteases:

- Main Protease: Mpro;
- Papain-Like Pro: PLpro;

Mpro = CLpro
Chymotripsin-like

Literature Review: CoV-2

PubMed Query:

- SARS CoV-2;
- Immuno-informatics or in-silico;

Refinement:

- Manually => Pharmacology, Vaccinology, Other;
- NO loss of any articles;

Literature Review: CoV-2 / in-silico

PubMed Articles:

	2020	2021	2022	2023 2025
Total	602	1,128	1,027	1,242
Vaccine	93	107	89	312
Pharma	380	450 *	276 *	ND
ND**	0	392 *	612 *	ND

Difference: Other categories (e.g. 129 in 2020);

* Mostly Pharma (NO vaccine); ** ND: Not yet done;

Literature Review: CoV-2 / in-silico

PubMed Articles:

	2020	2021
Total Vaccinology	**93**	**107**
Epitopes	**31**	**34**
Vaccines #	**51**	**64**
Other	**11 ***	**9 ***

2020 includes 2 Pre-Prints not indexed by PubMed;

only in-silico; * Reviews, Immunology (related to vaccines);

Literature Review: CoV-2 / in-silico

PubMed Articles:

Journals #	2020	2021
J Biomol Struct Dyn.	8	33
Sci Rep, Front Immunol, Comput Biol Med, Molecules	5 + 5 (2 + 1)	8 + 5 8 + 7
Vaccines (Bas.), J Med Virol, Inform Med Unlock, F1000Res	4 x 4	(more Ph.) 5, 3, 3, 1
Preprints $	5+	5+

in-silico Vacines & Epitopes; $ only PubMed (+ 1 other 2020);

Literature Review: CoV-2 / in-silico (only 2020)

Target Proteins:

- S: 24 (only S) + 15 (in combination);
- M: 16 (mostly in combination);
- Others: N (12), E, NSPs;

Epitope types:

- HLA-I, HLA-II;
- LB: linear & discontinuous;

Literature Review: CoV-2 / in-silico (only 2020)

CoV-2 Genomes:

- **Single strain:** possibly most; (*)
- Consensus sequence from **multiple strains**;
- Example PMID 32832263:
  - Consensus: 475 strains;
  - Epitopes: 13 HLA-I & 10 HLA-II;
  - Source: S, M and other proteins;

* Not yet quantified;

Literature Review: CoV-2 / in-silico

Vaccine Construct:

- CTL Epitopes;
- HTL Epitopes;
- B Epitopes;
- Adjuvants: beta-Defensin, PADRE, other sequences;
- Linkers: AAY, GPGPG, EAAAK, …;

Example:

- PMID: 32962156
- bDef – EAAAK – $(CTL/AAY)_{35}$ – $(HTL/GPGPG)_{21}$ – bDef
- Total: 1057 aa;

Literature Review: CoV-2 / in-silico

Docking:

- TLR: 3 & 4 most frequent (at least 23 & 14);
- Other: TLR 2,5,7,8,9 (a few);
- HLA molecules;
- Constructs: ~ 100 aa – 1057 aa; (some heroic dockings)
- Compare Hb: each chain 141 – 146 aa;

Literature Review: CoV-2 / in-silico

Docking:

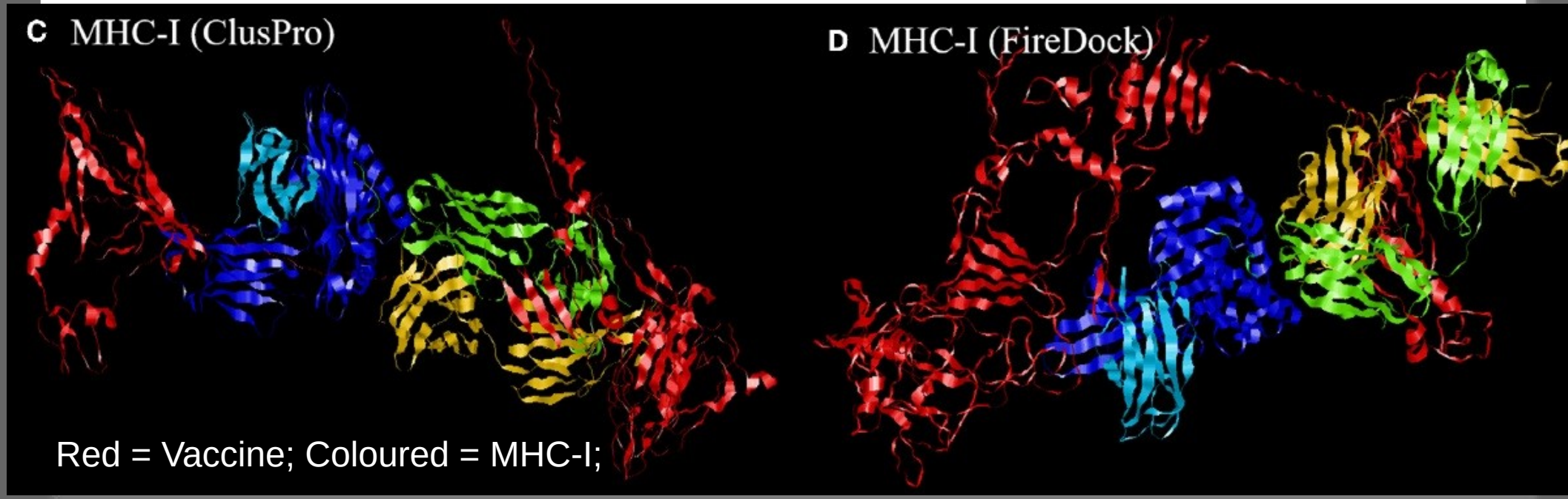
- TLRs & HLA;
- Constructs: ~ 100 aa – 1057 aa; (some **heroic dockings**)
- Compare Hb: each chain 141 – 146 aa;

Recommendations:

- **India & China** should probably prioritize development of infrastructures for *protein NMR;*
- Build first a solid foundation for protein structures;

Literature Review: CoV-2 / in-silico

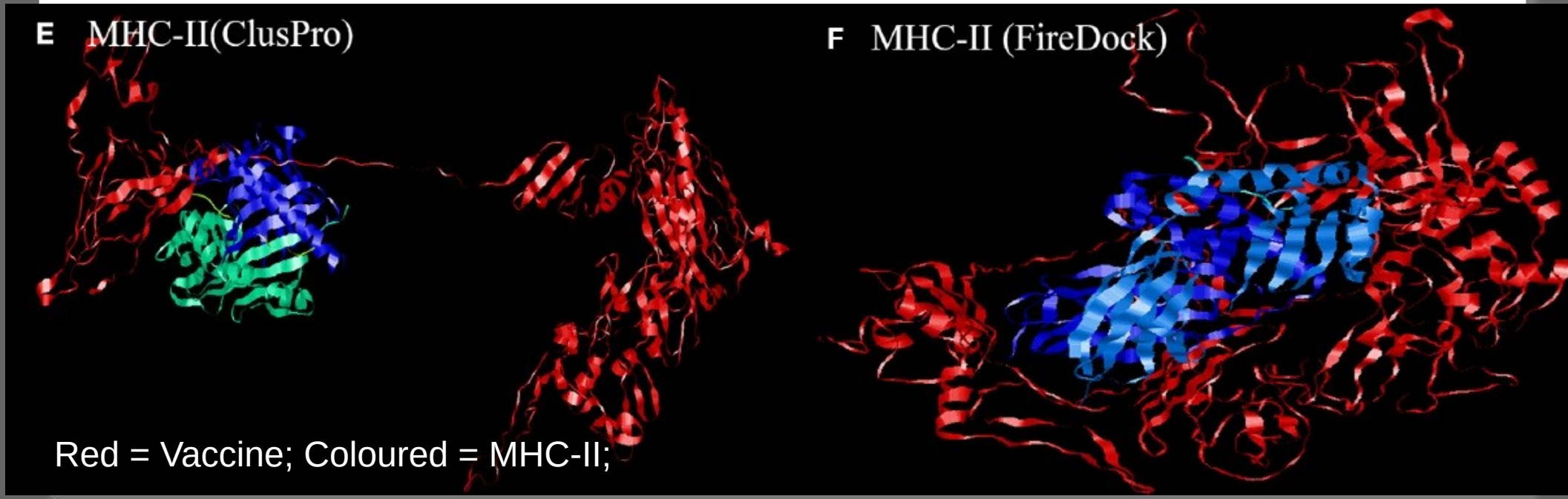
Docking: **MHC-I + whole protein Vaccine**



PMID: 32849643; Front. Immunol. 2020;

Literature Review: CoV-2 / in-silico

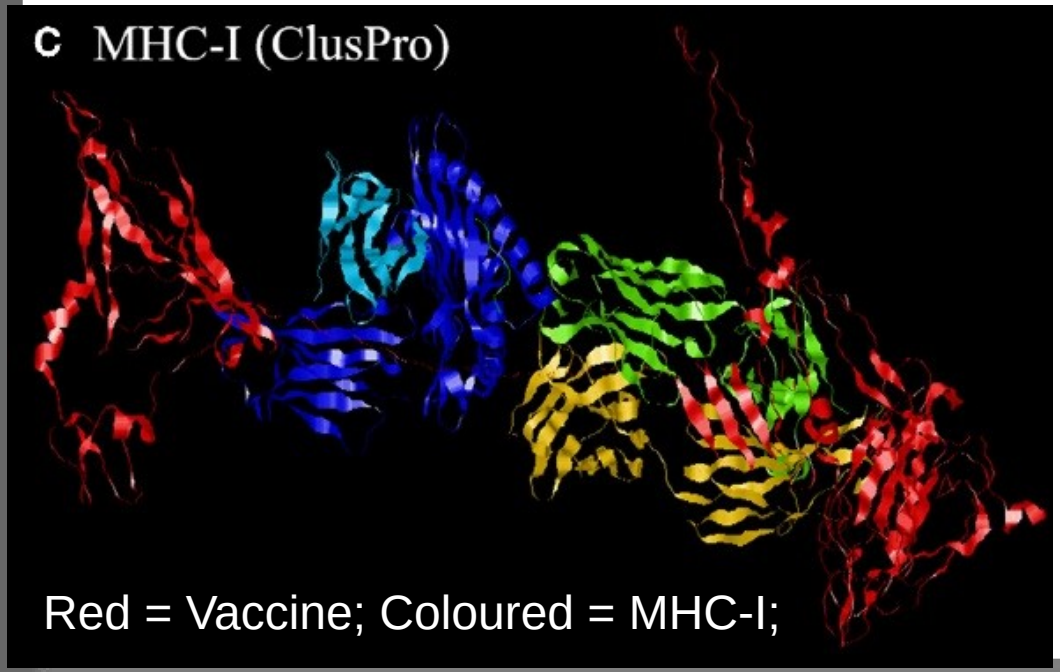
Docking: MHC-II + whole protein Vaccine



PMID: 32849643; Front. Immunol. 2020;

Literature Review: CoV-2 / in-silico

Docking: MHC-I + whole protein Vaccine



Is it possible to dock almost any protein?

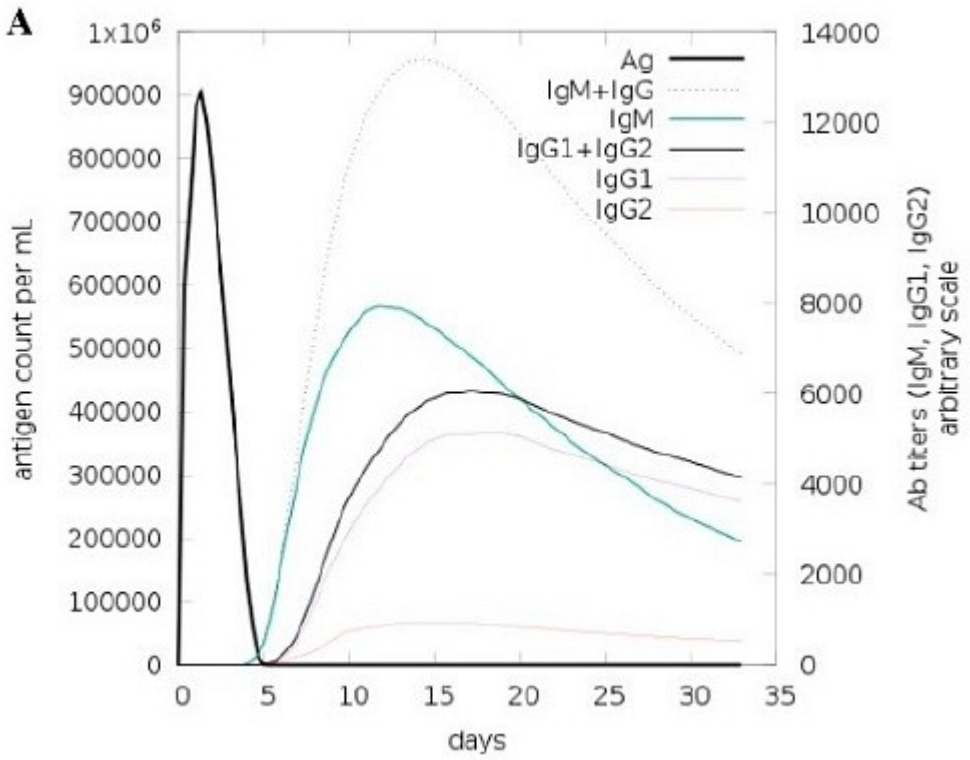
- Interesting docking exercise;
- Immunologically: absolute bollocks;
- Frontiers in Immunology!

PMID: 32849643; Front. Immunol. 2020;
For another example see: PMID 32616763; Sci Rep. 2020;

Literature Review: CoV-2 / in-silico

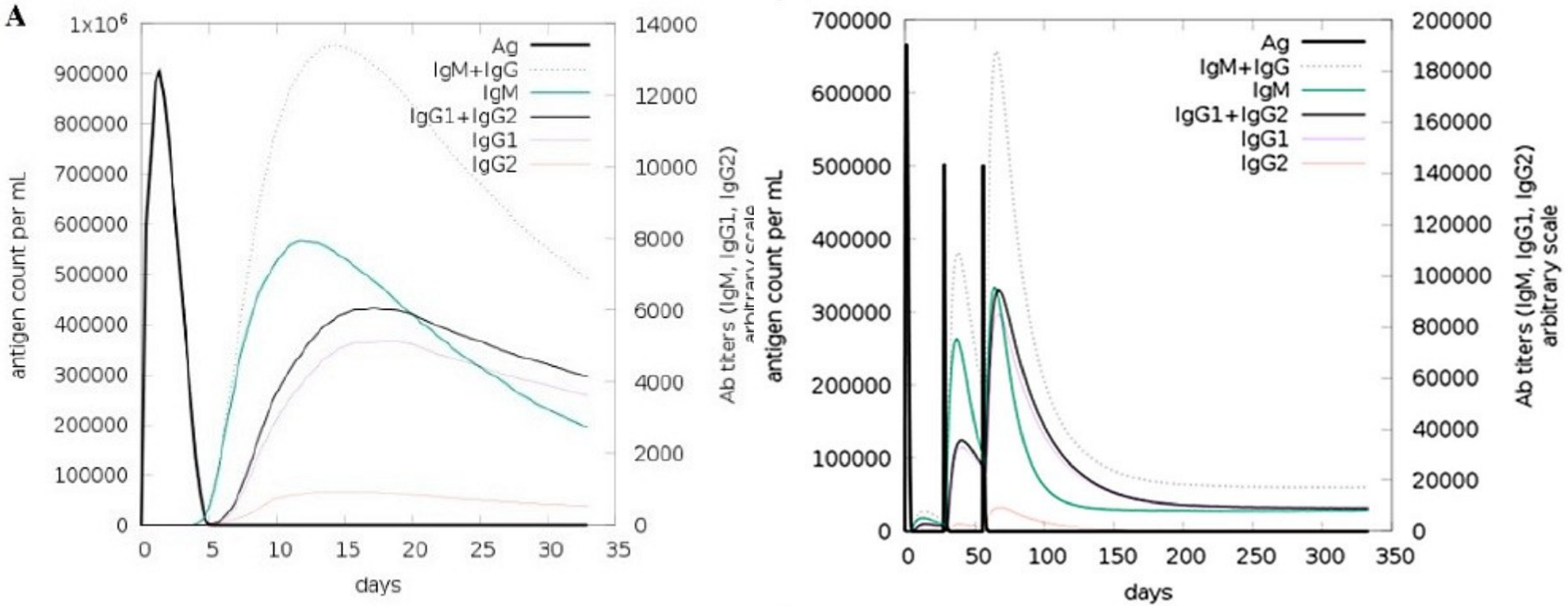
Immune Simulations:

- C-ImmSim 10.1 (2020)
- Ig sub-classes
- Cells: Tc, Th, LB
- Cytokines



Literature Review: CoV-2 / in-silico

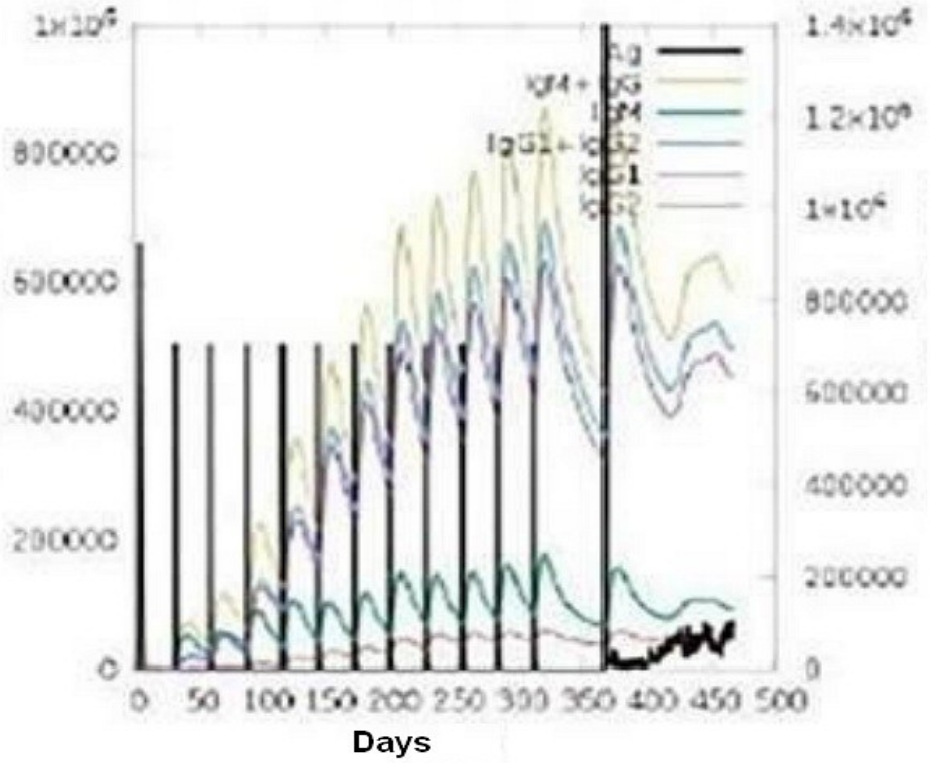
Immune Simulations: Antibodies



* Black = Antigen/Vaccine;

Literature Review: CoV-2 / in-silico

Immune Simulations: Antibodies after 12 x Vaccin!



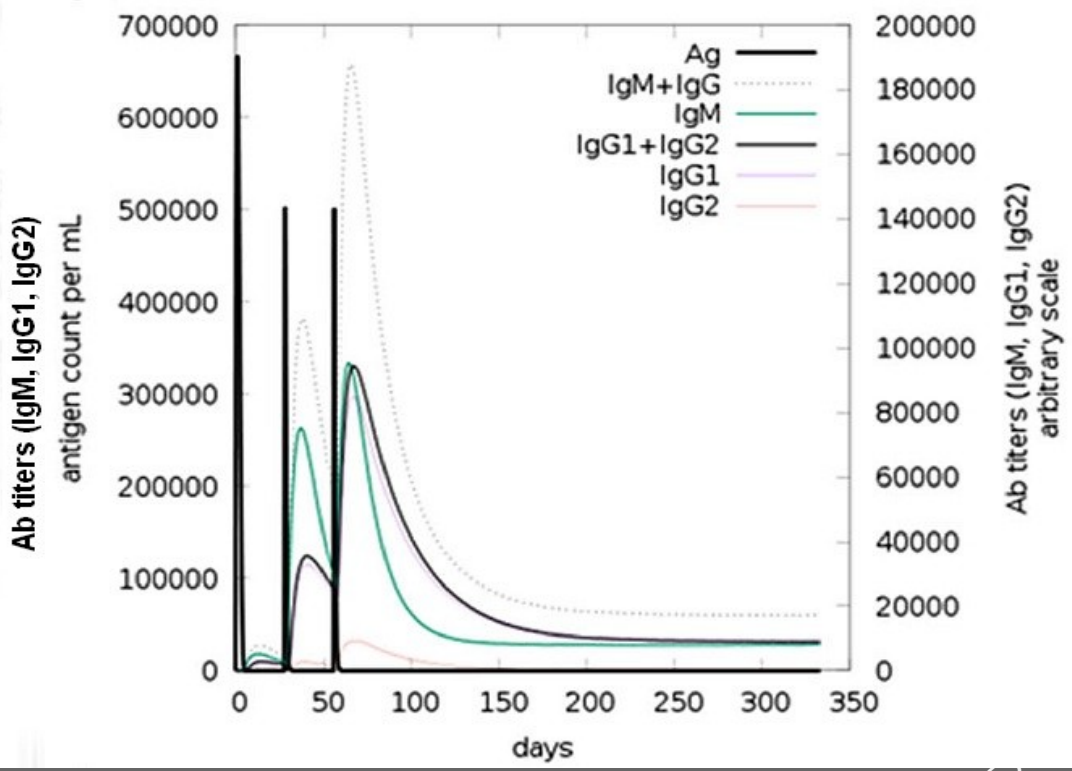
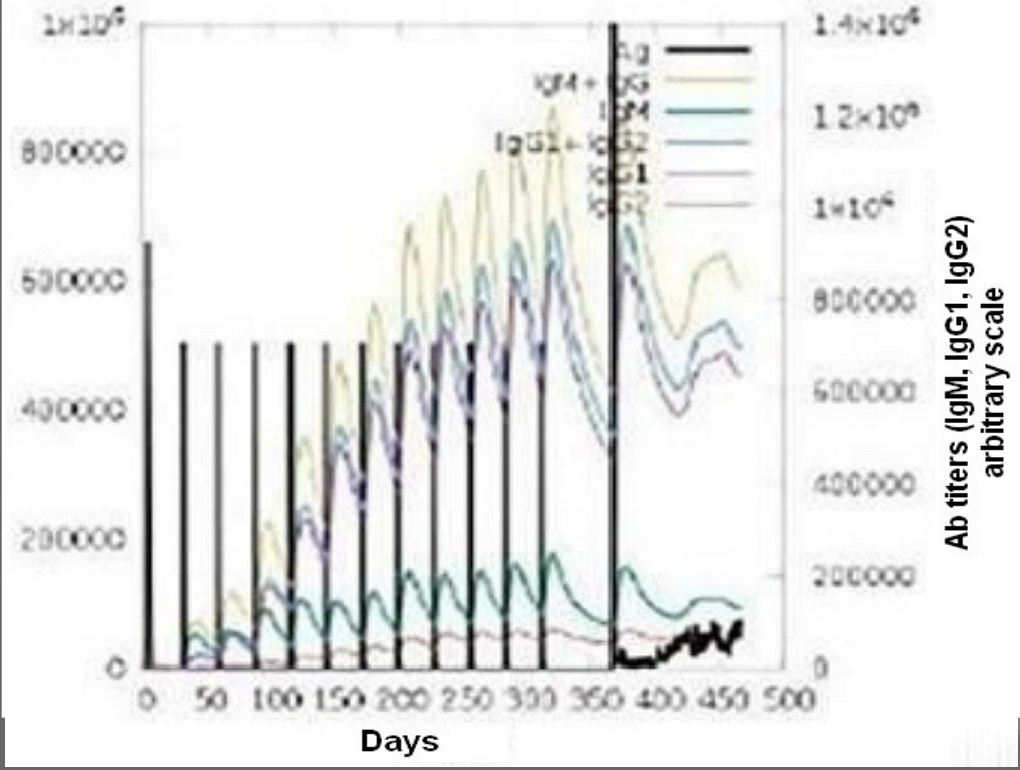


Fig 1 - Heroic Vaccination Strategy!

Literature Review: CoV-2 / in-silico

Immune Simulations: Antibodies after 12 x Vaccin!



Only Pre-print:

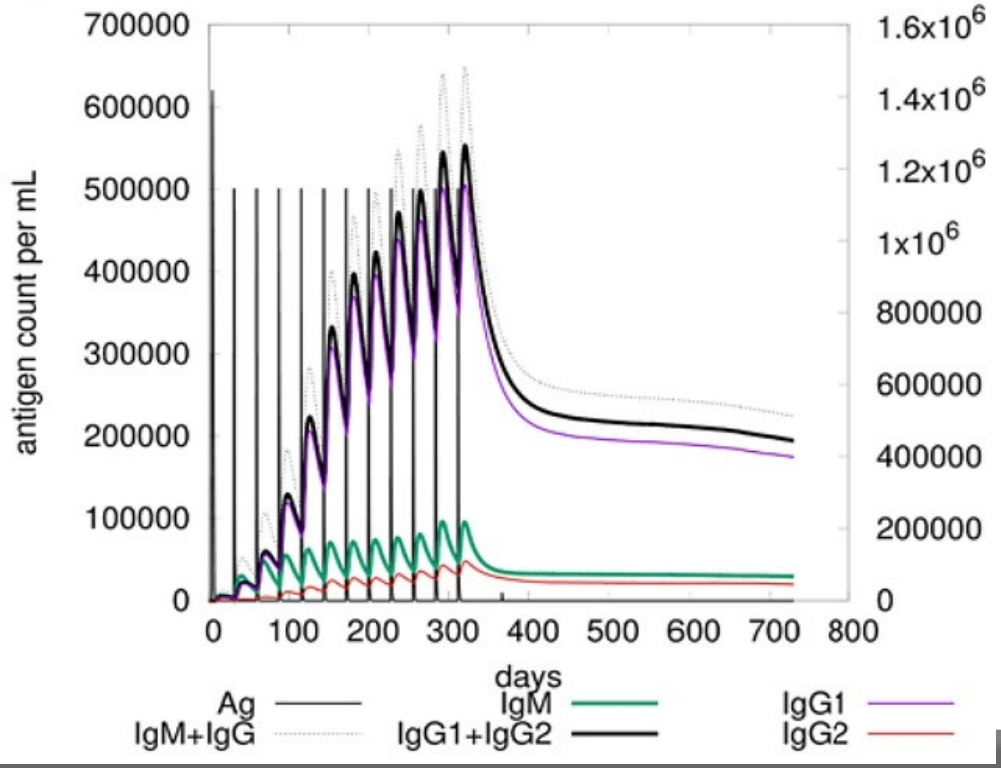
- 12 x Vaccine;
- Viral challenge afterwards;

Immuno-informatics approach for multi-epitope vaccine designing against SARS-CoV-2. bioRxiv 18-Aug-2020.

https://doi.org/10.1101/2020.07.23.218529;

Literature Review: CoV-2 / in-silico

Immune Simulations: Antibodies after 12 x Vaccin!



BUT this is published:

- 12 x Vaccine;
- *Viral challenge* afterwards;

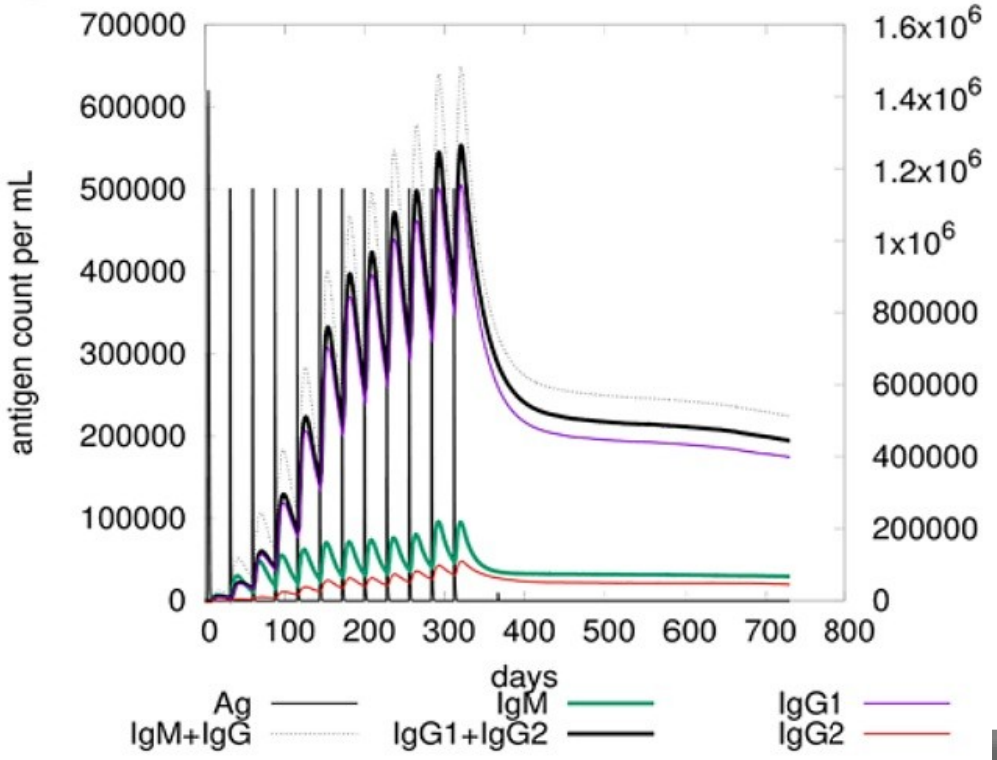
A candidate multi-epitope vaccine against SARS-CoV-2. Sci Rep. 2020 Jul 2; 10.

https://doi.org/10.1038/s41598-020-67749-1;

Viral Challenge: probably small tick at 365 days;

Literature Review: CoV-2 / in-silico

Immune Simulations: Antibodies after 12 x Vaccin!



BUT this is published:

- 12 x Vaccine;

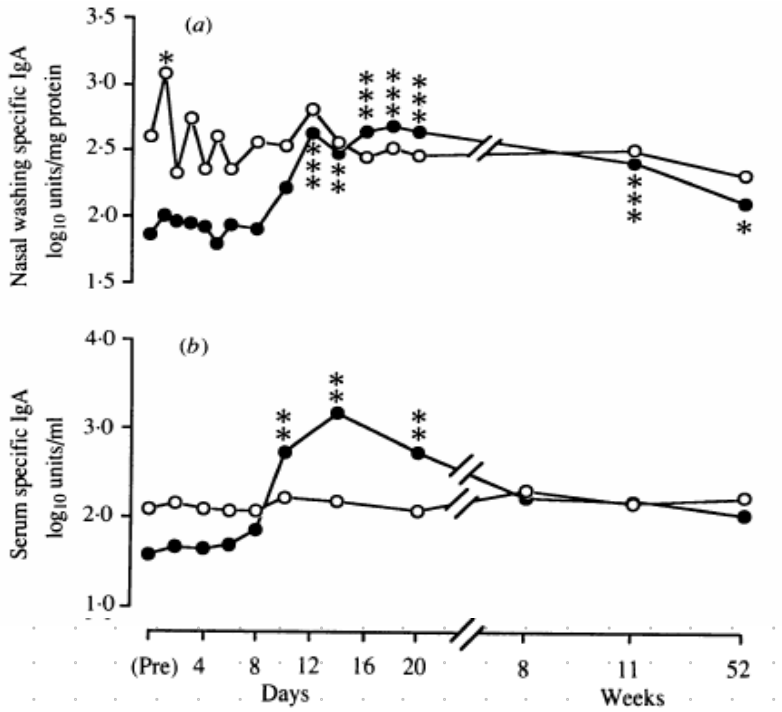
Recommendation:

Indian Gov can rehearse strategy to vaccinate each month the entire population for 12 consecutive months;

A candidate multi-epitope vaccine against SARS-CoV-2. Sci Rep. 2020 Jul 2; 10. https://doi.org/10.1038/s41598-020-67749-1;

Literature Review: CoV-2 / in-silico

Real IgA: Mucosa vs Serum



Volunteers inoculated with CoV 229E:

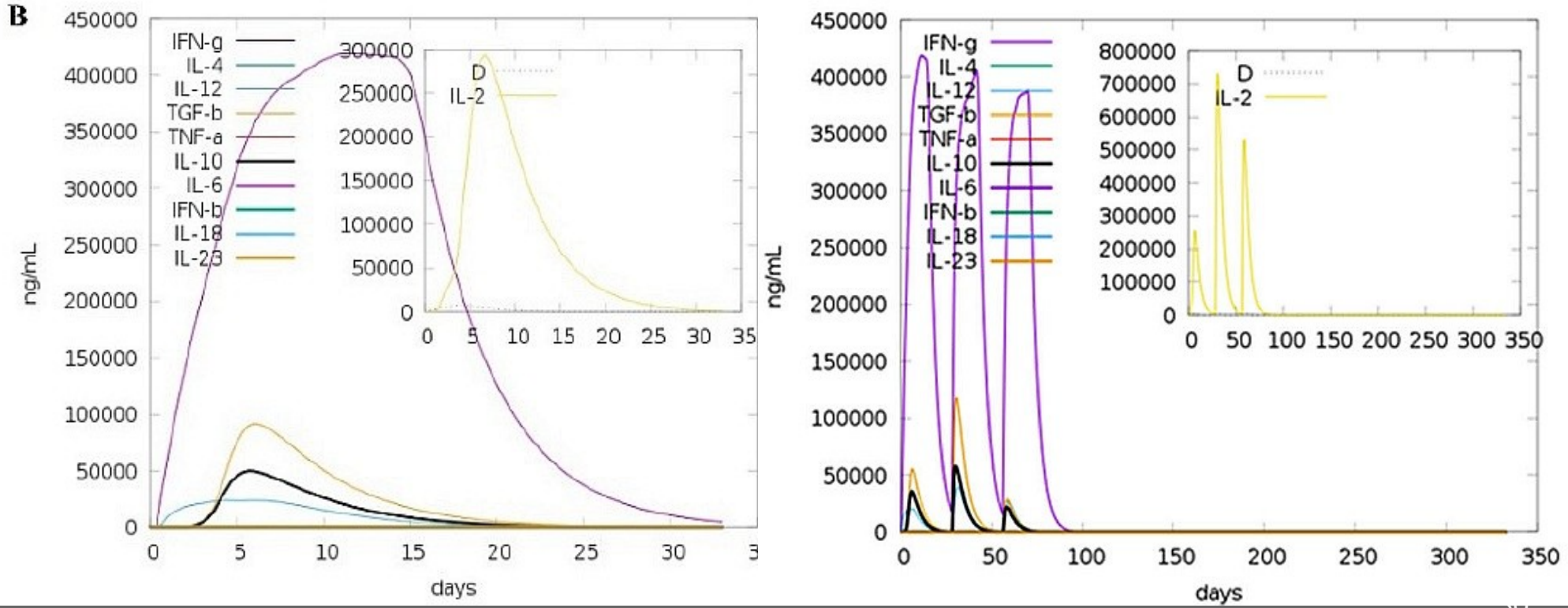
- Infected: 10;
- Uninfected: 5
  - Pre-existing **IgA**;
  - Surge in **Mucosal IgA**;

In-silico Literature:
NO simulations of IgA!

Fig. 2a,b: Callow KA et al. The time course of the immune response to experimental coronavirus infection of man. 1990. PMID 2170159.

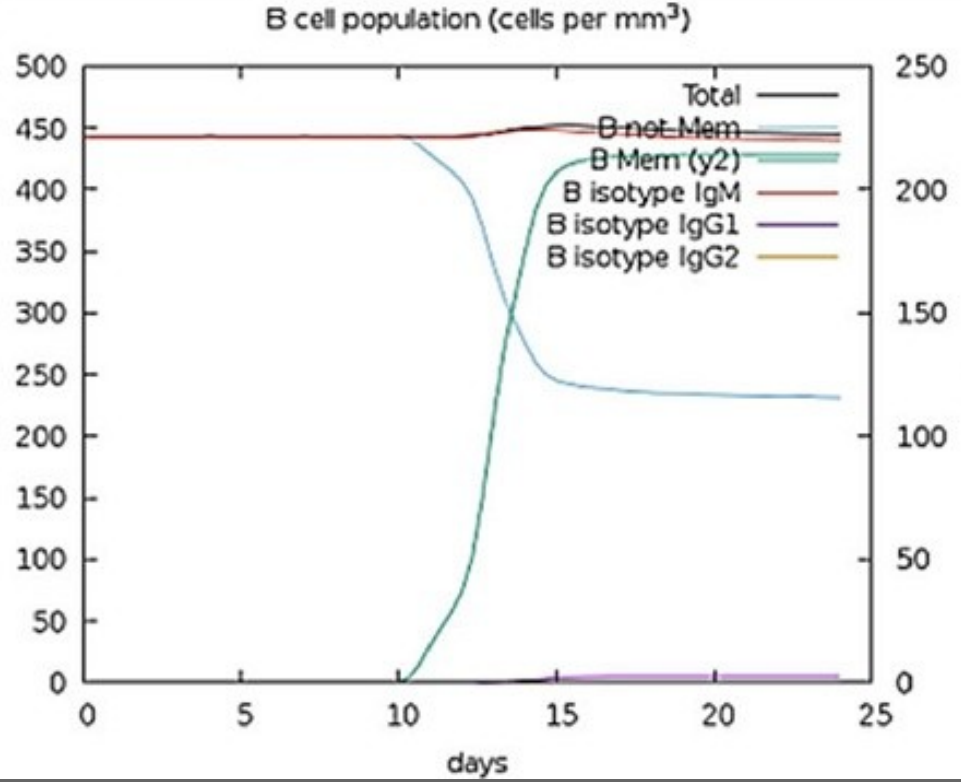
Literature Review: CoV-2 / in-silico

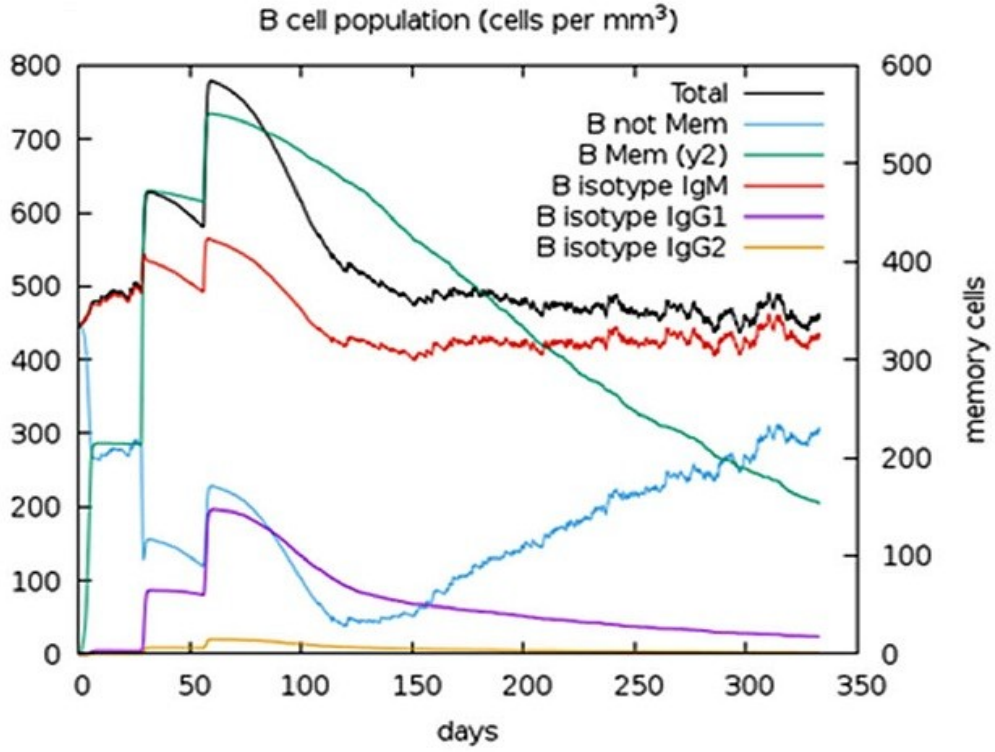
Immune Simulations: Cytokines



Literature Review: CoV-2 / in-silico

Immune Simulations: B Cells





TODO: Image with Vaccine at time = 0

Literature Review:

Recommendations: for PubMed & Journals

- **Mark articles with Caveat-Field:**
  - Docking Multi-Epitope vaccine with MHC-I or II: interesting docking exercise, but functionally completely irrelevant;
  - 12-Dose schedule: NO clinical utility;
  - Conformational Epitopes in Vaccine: **Neo-antigens** => **NO** protection against original virus;

* NO need for retraction: may still be useful exercises to learn Immunology or test in-silico methods / tools;

Pharmacology: CoV-2

- **Pharmacology:**

  Majority of PubMed articles with “**in-silico**”

Pharmacology

PubMed Articles:

- 2020: 339 + most of remaining 70;
- Drug-Repurposing & Drug-Libraries;
- Natural-/Phytocompounds;

Molecular Docking:

- Almost ALL articles;

Pharmacology

PubMed Articles:

- 2020: 339 + most of remaining 70;

Summary: based on ~ 50% of articles:

- Phytocompounds: 240; (*)
- Plants: 61; (*)
- Drugs: 140;

* Final number may be x 2;

Pharmacology: Plants

Plants & Phytocompunds:

- 2020: 339 + most of remaining 70;
- Phytocompounds: 240;
- **Plants:** 61;
- Drugs: 140;

* Summary: based on ~ 50% of articles; final number may be x 2;

Pharmacology: Phytocompunds

Plants & Phytocompunds:

- 2020: 339 + most of remaining 70;
- **Phytocompounds:** 240;
- Plants: 61;
- Drugs: 140;

* Summary: based on ~ 50% of articles; final number may be x 2;

Pharmacology: Targets

Targets:

- Main Protease (Mpro/CLpro): > 124;
- PL-Protease: alone or in combination (few);
- Polymerase (RdRp): > 11;
- S or S/ACE2 complex: > 45;
- Other: Methyltransferase, etc.